ClimateScope : Visualizing Global Weather Trends and Extreme Events

Milestone 2: Core Analysis & Visualization Design

1. Introduction

This document presents the outcomes of **Milestone 2** of the *ClimateScope* project.

The primary objective of this phase was to perform in-depth statistical and exploratory analysis on the processed global weather dataset, identify key climate patterns and extreme events, and design an interactive dashboard layout to effectively communicate insights.

This milestone builds upon the cleaned and aggregated dataset developed in Milestone 1 and focuses on analytical understanding rather than dashboard implementation.

2. Dataset Used

- **Source:** Global Weather Repository (Kaggle)
- **Data State:** Cleaned and processed (Milestone 1 output)
- **Granularity:** Monthly aggregated data
- **Key Variables Analyzed:**
  - Temperature (°C)
  - Humidity (%)
  - Precipitation (mm)
  - Wind Speed (kph)
  - Atmospheric Pressure (mb)
  - Season, Country, Year, Month

3. Statistical & Distribution Analysis

Descriptive statistics and distribution plots were generated to understand the behavior of core weather variables.

Key Observations:

- Temperature values fall within realistic global climatic ranges.
- Precipitation shows a right-skewed distribution with frequent low-rainfall observations.
- Wind speed and humidity distributions align with expected meteorological patterns.

These findings confirm the reliability and suitability of the dataset for further climate analysis.

4. Correlation Analysis

A correlation matrix and scatter plots were used to explore relationships between major weather variables.

Key Findings:

- Temperature and humidity show a weak to moderate correlation depending on region.
- Atmospheric pressure exhibits an inverse relationship with wind speed in some cases.
- Precipitation shows limited direct linear correlation with temperature, indicating the influence of additional climatic factors.

This analysis highlights the complex, non-linear nature of climate interactions.

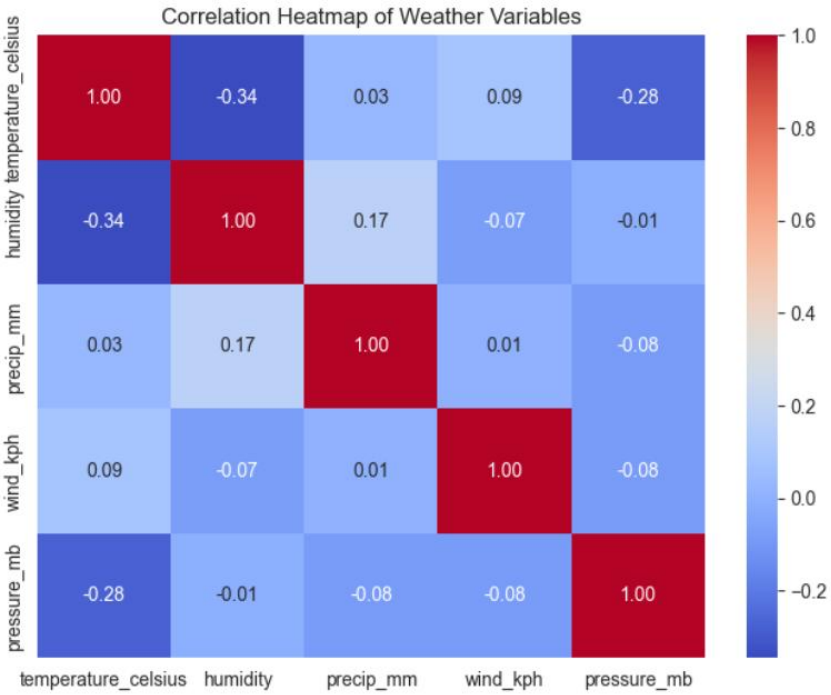

Figure 1: Correlation Heatmap of Weather Variables

Figure 1 illustrates the correlation heatmap of key weather variables including temperature, humidity, precipitation, wind speed, and atmospheric pressure. The visualization highlights both positive and negative linear relationships among variables. Temperature and humidity exhibit a weak to moderate correlation, while wind speed shows an inverse relationship with atmospheric pressure in some cases. This heatmap provides a compact statistical overview of inter-variable dependencies and helps identify which weather parameters are more closely related.

5. Seasonal Weather Patterns

Seasonal analysis was performed using derived season labels (Winter, Spring, Summer, Autumn).

Key Findings:

- Clear seasonal temperature variation is observed globally.
- Summer consistently records the highest average temperatures.
- Winter shows the lowest average temperatures across most regions.

- Seasonal precipitation patterns vary significantly between countries.

This confirms the effectiveness of seasonal classification and aggregation logic.

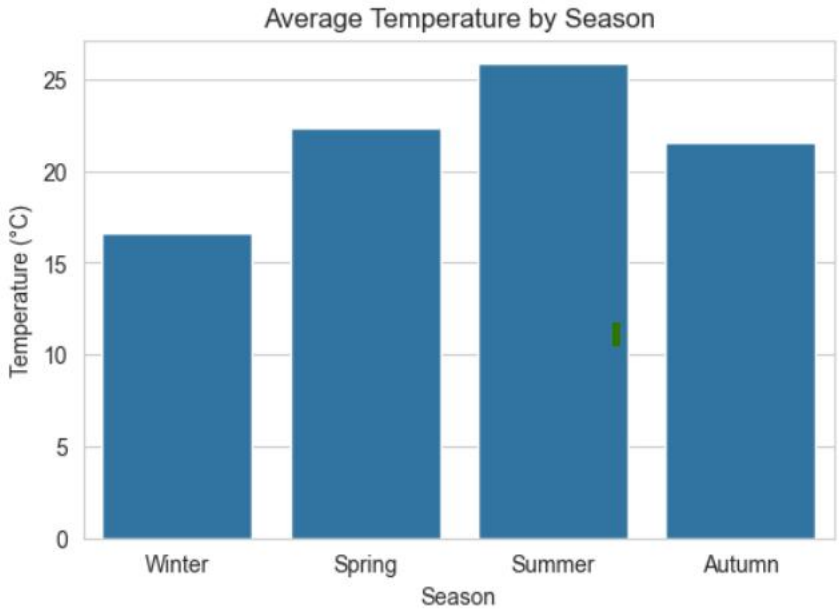

Figure 2: Average Seasonal Temperature by Season

Figure 2 shows the average temperature variation across meteorological seasons (Winter, Spring, Summer, and Autumn). The visualization clearly demonstrates seasonal temperature patterns, with summer months consistently recording higher average temperatures and winter months showing lower averages. This confirms the effectiveness of seasonal classification and validates the presence of expected climatic seasonality in the dataset.

6. Regional Comparison

Weather conditions were compared across selected countries and regions to understand geographic variability.

Key Findings:

- Significant differences exist in average temperature and wind speed across regions.
- Some regions exhibit consistently higher wind speeds, indicating geographic or coastal influence.
- Regional comparisons emphasize the diversity of global climate behavior.

7. Extreme Weather Event Identification

Extreme events were defined as observations exceeding the **95th percentile** of temperature or precipitation.

Key Findings:

- Extreme temperature and rainfall events are relatively rare but concentrated in specific regions.
- Scatter plot visualization highlights anomalies that deviate significantly from normal climate behavior.
- This method provides a simple and interpretable approach to identifying climate extremes.

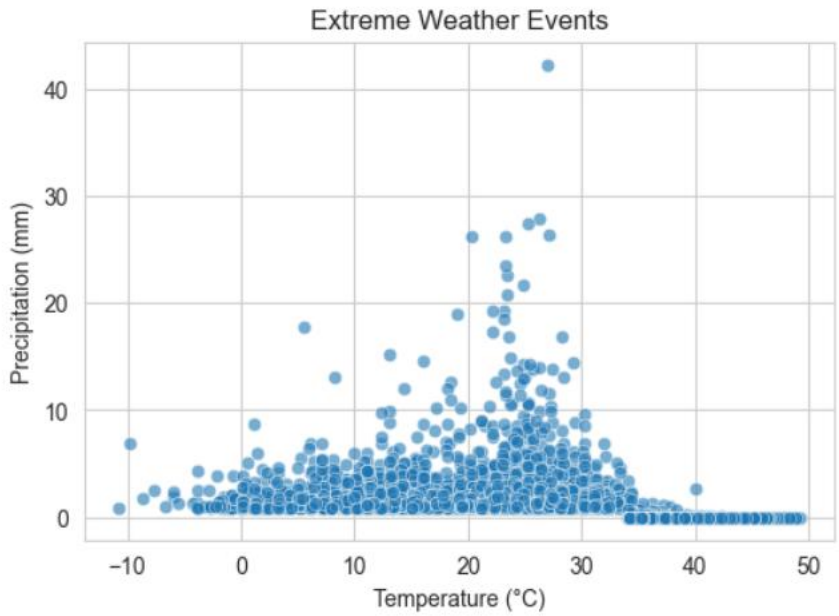

Figure 3: Extreme Weather Events Distribution

Figure 3 visualizes extreme weather events identified using a percentile-based approach, where extreme temperature or precipitation values exceed the 95th percentile. The scatter plot highlights anomalous observations that significantly deviate from typical climate behavior. These extreme events are relatively infrequent but appear clustered in specific regions, indicating areas more prone to climate extremes.

8. Trend Analysis

Monthly average temperature trends were analyzed to assess temporal changes.

Key Findings:

- Clear seasonal oscillations are present in temperature trends.
- Subtle long-term warming signals are observable in aggregated trends.
- These patterns support the importance of long-term climate monitoring.

9. Dashboard Design & Wireframe

A multi-section dashboard wireframe was designed to visualize the analytical findings.

Dashboard Sections:

1. **Global Overview**
    - **Global temperature trend line**
    - **KPI cards for average temperature, humidity, and rainfall**
2. **Seasonal Analysis**
    - **Country selector**
    - **Seasonal temperature and precipitation charts**
3. **Regional Comparison**
    - **Choropleth map for average temperature**
    - **Wind speed comparison by region**
4. **Extreme Weather Events**
    - **Scatter plot of extreme temperature and precipitation events**
    - **Filters for year and country**

The wireframe illustrates chart placement, filters, and user interaction flow, serving as a blueprint for dashboard implementation in the next milestone.The complete dashboard wireframe is provided as a separate document:

File: reports/dashboard_wireframe.pdf

This wireframe illustrates chart placement, filters, and user interaction
flow for global overview, seasonal analysis, regional comparison, and
extreme event detection.

Conclusion

Milestone 2 was successfully completed with comprehensive statistical analysis, seasonal and regional climate exploration, extreme event identification, and trend assessment.

The designed dashboard wireframe effectively translates analytical insights into an intuitive visualization plan. This milestone establishes a strong analytical and design foundation for implementing an interactive climate dashboard in the next phase.

Next Steps

- Implement the dashboard using Streamlit and Plotly
- Add interactive filters and dynamic visualizations
- Integrate processed data into the dashboard backend
- Enhance user experience and visual consistency